STINK3024 Machine Learning
Session II 2025/2026 (A252)
Project Machine Learning Development and Reporting (20%)
LOC 3a (20%)

Type: Group work
Submission: Week 13 (18 June 2026)

1. **Describe the machine learning model development process** clearly, including data preparation, feature engineering, algorithm selection, and parameter tuning.
2. **Present and justify the chosen model architecture or algorithm**, supported by relevant theoretical background and practical reasoning.
3. **Interpret and explain model performance results** using appropriate evaluation metrics (e.g., accuracy, precision, recall, F1-score, ROC-AUC).
4. **Compare and discuss alternative models or configurations**, highlighting strengths, weaknesses, and trade-offs in performance.
5. **Visualize and communicate analytical results effectively** through tables, graphs, and other visual aids.
6. **Demonstrate the ability to write a structured technical report**, incorporating an introduction, methodology, results, discussion, and conclusion.
7. **Reflect on limitations, challenges, and potential improvements** to the developed model and future research directions.

Term and Condition: a penalty of 2 marks will be applied for plagiarism or late submission
Objective: To expose student to the process of machine learning development and reporting
Detail Report Structure (Target 40 – 50 pages)
Report format: Times New Roman, 1.5 spacing and 12 font size

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1. Title Page (1 page)

- Title of the report
- Student name
- Institution/organization
- Course
- Date of submission

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2. Declaration & Acknowledgements (1 page)

- Short declaration of originality
- Acknowledgement of guidance, dataset providers, tools, etc.

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3. Abstract (1 page)

- Concise summary of the project's goals, methods, key results, and conclusions

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4. Table of Contents (1 page)

- Auto-generated TOC with section numbers and page numbers

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5. List of Figures & Tables (1 page)

- Auto-generated if using Word or LaTeX

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6. Introduction (3–4 pages)

- Background of the problem
- Motivation and relevance
- Real-world applications and potential impact
- Report objectives and research questions
- Overview of the methodology and report structure

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7. Related Work (6–8 pages)

- Review of academic papers, industry applications, and benchmarks
- Summary of machine learning techniques used previously for similar problems
- Discussion on the limitations and gaps in existing approaches
- Comparison table for related works
- How your work differs/improves/extends previous work

Tips:

- Use scholarly sources (e.g., Google Scholar, IEEE Xplore)
- Include citations and a comparative summary table

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8. Dataset and Problem Definition (4–5 pages)

- Detailed description of dataset(s) used
- Features and target variable(s)
- Class distribution or data characteristics (EDA summaries)
- Specific ML problem definition (classification, regression, etc.)
- Challenges (missing data, imbalance, noise, etc.)

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9. Methodology (8–10 pages)

- Preprocessing steps (cleaning, imputation, encoding, normalization)
- Feature engineering/selection techniques
- Algorithms chosen and their mathematical foundations (briefly)
- Reason for algorithm selection (theoretical or empirical basis)
- Train/test split, validation techniques (k-fold, stratified sampling, etc.)
- Hyperparameter tuning strategies (GridSearchCV, RandomizedSearch, Bayesian Optimization)
- Tools used (e.g., Python, Scikit-learn, XGBoost, TensorFlow)

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10. Results and Evaluation (6–8 pages)

- Summary tables of model performance (accuracy, F1-score, AUC, etc.)
- Visualizations (confusion matrices, ROC curves, residual plots, etc.)
- Learning curves, validation vs training performance
- Feature importance plots or SHAP values
- Explanation of model behavior or misclassifications

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11. Discussion (4–6 pages)

- Analysis of which models performed best and why
- Interpretation of evaluation metrics in context
- Cross-reference with literature: how do your results compare?
- Model robustness and generalizability
- Potential biases, shortcomings, or anomalies
- Relevance to real-world deployment

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12. Conclusion and Future Work (2–3 pages)

- Summary of findings and their implications
- Final model recommendations

- What could be improved or extended (e.g., data quality, deep learning, larger samples)
- Ideas for deployment, productization, or academic continuation

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13. References (2–3 pages)

- All cited sources using a consistent style (APA, IEEE, etc.)
- Include papers, books, datasets, libraries/tools documentation

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14. Appendices (5–8 pages)

- Full code snippets or repository link
- Extended tables or charts
- Parameter tuning grids or logs
- Raw evaluation outputs
- Ethics form (if applicable)

Tips to Reach 40–50 Pages Naturally

- Use detailed analysis and plenty of visualizations (charts, graphs, tables)
- Include multiple models and tuning stages (baseline + advanced)
- Expand literature review with 10–15 well-cited references
- Add insights from interpretability tools like SHAP/LIME
- Include technical theory in small, digestible portions (especially in the methodology)